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Magnetostratigraphy and high-resolution lithostratigraphy of the Permian–Triassic boundary interval in Central Germany

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Abstract

A comprehensive lithostratigraphic and magnetostratigraphic study was carried out in order to provide a magnetic polarity scale for the Lower Buntsandstein of the classic Germanic Trias. In its type area of Central Germany the ~300 m thick Lower Buntsandstein consists of 20 small-scale fining-upward cycles. They can be correlated over large parts of the Central European Basin using a combination of high-resolution cyclic stratigraphy and gamma-ray logging. On the basis of this robust lithostratigraphic framework a very precise positioning and correlation of paleomagnetic results has been realized. Eight outcrops and one completely cored well were collected, yielding a total of ~900 paleomagnetic samples. From about 80% of them a characteristic remanence was obtained. The consistent correlation of lithostratigraphic units and magnetozones supports an early acquisition of the characteristic magnetizations in both the magnetite-bearing gray lithologies as well as the hematite-bearing red-brown lithologies. The inter-section correlation of all investigated profiles allows the creation of a well-defined composite magnetic polarity scale. The uppermost Zechstein comprises one normal (znz) and two reversed (zry, zrz) magnetozones. In the overlying Lower Buntsandstein are delineated three normal (sn1 to sn3) and three reversed (sr1 to sr3) well-documented magnetozones, which can be correlated with magnetic records from the Boreal and Tethyan realms. The combination of magnetostratigraphic and biostratigraphic data suggests the base of the Oolite Alpha 2 within the beds of the so-called 'Graubankbereich' (Lower Buntsandstein) as the position of the *Hindeodus parvus* calibrated Permian–Triassic boundary within the Germanic Trias. It is situated within the lowermost part of a remarkable thick normal magnetozone (sn1), which is a distinctive feature occurring in virtually all magnetic records across the Permian–Triassic boundary.
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1. Introduction

The Buntsandstein represents the lower group of the tripartite classic Germanic Trias [1]. It is traditionally subdivided by lithological criteria

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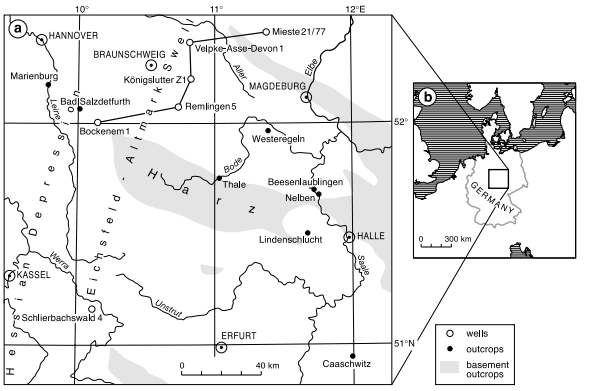


Fig. 1. (a) Simplified map of Central Germany with the location of the outcrops and wells investigated. In addition, the NNE–SSW-trending paleotectonical elements Hessian Depression and Eichsfeld–Altmark Swell are shown. Location of wells used for GR log cross-section (see also Fig. 4). (b) Position of map shown in panel a.

into three subgroups: (a) the fine-grained Lower Buntsandstein, comprising characteristic oolites, (b) the Middle Buntsandstein composed of coarse-grained sandstones and shales, and (c) the shaly Upper Buntsandstein containing evaporites. Traditionally, the lithostratigraphic Zechstein–Buntsandstein boundary was compared with the ammonite and conodont calibrated Permian–Triassic boundary (PTB; e.g. [2]). By decision of the IUGS (March 2001 [3]) the Global Stratotype Section and Point of the PTB is now the Meishan section D (South China), base of Bed 27c, at the first appearance of the conodont *Hindeodus parvus*.

The global nature of geomagnetic polarity reversals has made magnetostratigraphy an essential tool for precise correlation between sections of rocks, of different lithofacies and different depositional environments. Since the PTB Working Group was established, significant progress has been made in constructing a magnetic polarity scale for the Permian–Triassic boundary interval. Recent studies concerning the PTB were carried out in Canada [4], Svalbard [5,6], Italy [7], Transcaucasia [8], Iran [9], Pakistan [10] and China [11–17]. In the last four decades paleomagnetic research in the Buntsandstein focused on the relatively coarse-grained marginal facies of South Germany [18–21]. Until this study, it has been difficult to exactly compare this facies with the well-established high-resolution lithostratigraphy of the fine-grained basinal type area in Central Germany. Moreover, significant time gaps are present in the marginal succession, due to discontinuous sedimentation. Besides, it seems that the marginal facies is less suitable as a paleomagnetic recorder, because of widespread remagnetization.

The aim of this study is to establish a composite magnetic polarity scale for the basinal facies of the Lower Buntsandstein in its type area of Central Germany. In the surroundings of the Harz Mountains nine sections (eight outcrops, one well) were investigated using both lithostratigraphic and magnetostratigraphic tools (Fig. 1). Combining high-resolution cyclic stratigraphy and gamma-ray logs (GR logs), a precise positioning of paleomagnetic results has been realized [22]. Gamma-ray logging of outcrops offers a promising technique to directly compare well data with surface geology [23], particularly in monotonous fine-grained successions like the Lower Buntsandstein. Using a combination of magnetostratigraphic and biostratigraphic data, this study contributes to an improved correlation of the Lower Buntsandstein with the Triassic stages.

2. Geological setting

In Central Germany, the Lower Buntsandstein is ~300 m thick and consists of red beds, predominantly fine-grained sandstones, siltstones and shales with interbedded oolitic limestones. The clastic sediments were deposited during Latest Permian to Early Triassic times in fluvio-lacustrine environments of a large intracratonic basin [24]. The oolites were formed in hypersaline playa lakes. The succession is poor in fossils. Therefore, its subdivision and correlation are based on lithological criteria (Fig. 2). The Lower Buntsandstein is subdivided into the Calvörde Formation and the Bernburg Formation [28]. Its lower boundary is placed at the base of the first distinct sandstone interval above the Zechstein. The upper boundary is marked by the Volpriehausen unconformity at the base of the Volpriehausen Formation, which is characterized by the first basinwide

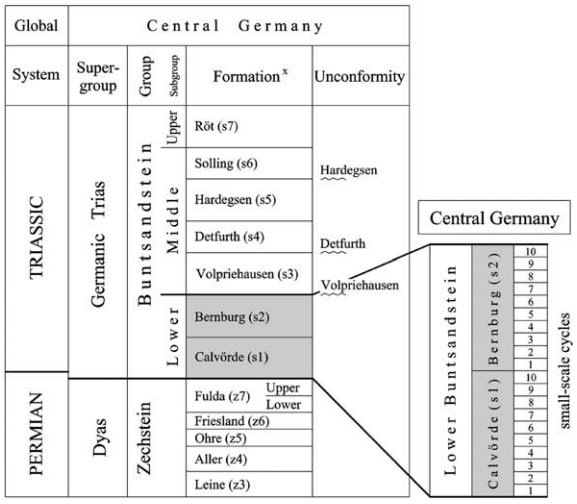


Fig. 2. Lithostratigraphy of the upper Zechstein and Buntsandstein in Central Germany with Calvörde and Bernburg formations, both consisting of 10 small-scale fining-upward cycles [22]; [x] 'Folgen' sensu Menning [25], Käding [26] and Stratigraphic Table of Germany 2002 [27].

influx of coarse-grained sands (Volpriehausen Sandstone Member) above the fine-grained Lower Buntsandstein.

The clastic–carbonatic succession shows a distinct cyclicity of varying magnitude, pragmatically subdivided into 20 approximately 10–25 m thick small-scale fining-upward cycles (so-called 'Kleinzyklen'). Commonly, these cycles begin with thin sandstones that grade into more pelitic sediments. Oolitic limestones (so-called 'Rogensteine') occur above the lower sandy part. The cycles can be used for correlation in wireline logs, cores and outcrops. With wireline logs (e.g. GR logs) they can be traced over almost the entire Central European Basin (CEB) from the Netherlands in the west via Germany to Poland in the east [22,29–32]. The oolites form correlatable litho-markers, indicating a large lateral consistency of facies development. Additionally, several other prominent litho-markers support the correlation, for example the 'Graubankbereich' (= gray bed interval). Alternatively, the small-scale fining-upward cycles can be interpreted as more or less symmetrical base-level cycles [22]. They are considered to reflect climatic fluctuation of alternating drier and wetter periods due to solar-induced ~100 ka Milankovitch eccentricity cycles [22,31,33]. In terms of sequence stratigraphy the Lower Buntsandstein is considered to represent the first third-order sequence of the Germanic Trias [22,34].

The biostratigraphic subdivision of the Buntsandstein is based mainly on conchostracans and sporomorphs [35–39]. The Lower Buntsandstein can be subdivided into nine conchostracan zones [39]. Also, they have been found in brackish intercalations of shallow marine deposits in Hungary, enabling the direct correlation of marine and continental biozones [40]. Thus, based on conchostracans the PTB has been placed within the 'Graubankbereich' (cycle 2, Calvörde Formation), where the Permian *Falsisca postera* is replaced by the Triassic *F. verchojanica* [39].

3. High-resolution lithostratigraphy

3.1. Gamma-ray logging

The natural gamma radiation of eight outcrops was recorded using a portable scintillation counter designed by G.B.-H. Elektronik [23]. The sections were measured every 10 cm over a selected time integration of 10 s, resulting in a higher resolution than the wireline gamma logs. The uncalibrated measurements are stored in the unit as counts per second (cps). The GR logs of the Buntsandstein sections basically display the content of ^{40}K, which is present in clay minerals. Therefore, the logs reflect more or less the clay content of the clastic rocks. With the help of gamma-ray logging it is possible to define gamma markers and to directly compare the outcrops with the wireline logs from wells. Moreover, the GR logs allow us to correlate the small-scale cycles over large parts of the CEB.

3.2. Results of high-resolution lithostratigraphy and gamma-ray logging

The method of cyclic stratigraphy is exemplified in the 140 m thick Marienburg section, which comprises almost the entire Bernburg Formation and the lowermost Volpriehausen Formation (Fig. 3). It contains ten 10–25 m thick, small-scale

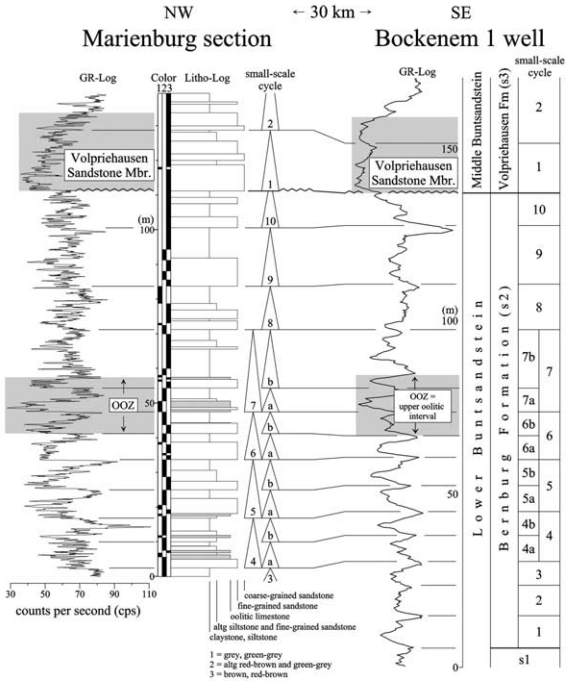


Fig. 3. GR log correlation between Marienburg section and Bockenem 1 well (from [41]).

fining-upward cycles. Locally, some of them can be subdivided into two subcycles. The cycles are obvious in the GR logs, with the sandy–oolitic part being indicated by very low radiation (30–60 cps) and the shaly upper part with high values (70–110 cps). In general, the sandstones are characterized by values between 40 and 60 cps, whereas the siltstones and shales vary between 60 and 120 cps. The thicker oolitic limestones have less than 30 cps, serving as good gamma markers. The GR log from the Marienburg section can readily be compared with the wireline gamma log of the Bockenem 1 well (Fig. 3). The so-called 'Obere Oolithische Zone' (= upper oolitic interval, OOZ), comprising cycles 6b to 7b, is characterized by regionally widespread thick oolite beds with low radiation in the GR log. The thick, coarse-grained sandstones at the base of the Volpriehausen Formation, the so-called 'Volpriehausen Sandstein' (Volpriehausen Sandstone Member), form another prominent marker horizon. Its base marks the Lower–Middle Buntsandstein boundary and is indicated by a distinct decrease in radiation from more than 80 cps to less than 30 cps.

The small-scale cycles provide a robust high-resolution lithostratigraphic framework in large parts of the CEB. Based on regional log correlations there is no evidence for hiatuses within the basinal facies of the Lower Buntsandstein [22,31]. Fig. 4 shows a 110 km long SW–NE-trending log transect crossing the paleogeographical elements of the Hessian Depression and the Eichsfeld–Altmark Swell. The seven other studied outcrops can be correlated with the same high reliability [22]. Thus, with a combined litho- and log-stratigraphic study, a precise positioning of all paleomagnetic results has been realized. Further details are given in Szurlies [22,32].

4. Rock magnetism and paleomagnetism

4.1. Materials and methods

In general, fine-grained sandstones and oolitic limestones were collected at 1–2 m intervals, yielding a total of nearly 900 oriented standard paleomagnetic samples. Whereas red-brown and brown sandstones (red-brown lithologies) were subjected to thermal demagnetization, gray-brown sandstones, gray calcareous sandstones and gray limestones (gray lithologies) were predominantly demagnetized by alternating fields. One subset of the red-brown lithologies was demagnetized using a MAVACS (Magnetic Vacuum Control System, AGICO) compensated oven. Measurements of natural remanent magnetization (NRM) were made with a JR5A spinner magnetometer (AGICO). The remainder was demagnetized with a shielded MMTD1 oven (Magnetic Measurements) with the NRM being measured using an automatic DC-SQUID 755SRM cryogenic magnetometer (2G Enterprises). The magnetic susceptibility was recorded after each demagnetization step with an AGICO Kappabridge KLY-2 in order to monitor mineralogical changes due to heating. The alternating field demagnetization of the gray lithologies was performed using the in-line triaxial alternating field demagnetizer of the cryogenic magnetometer. Isothermal remanent magnetiza-

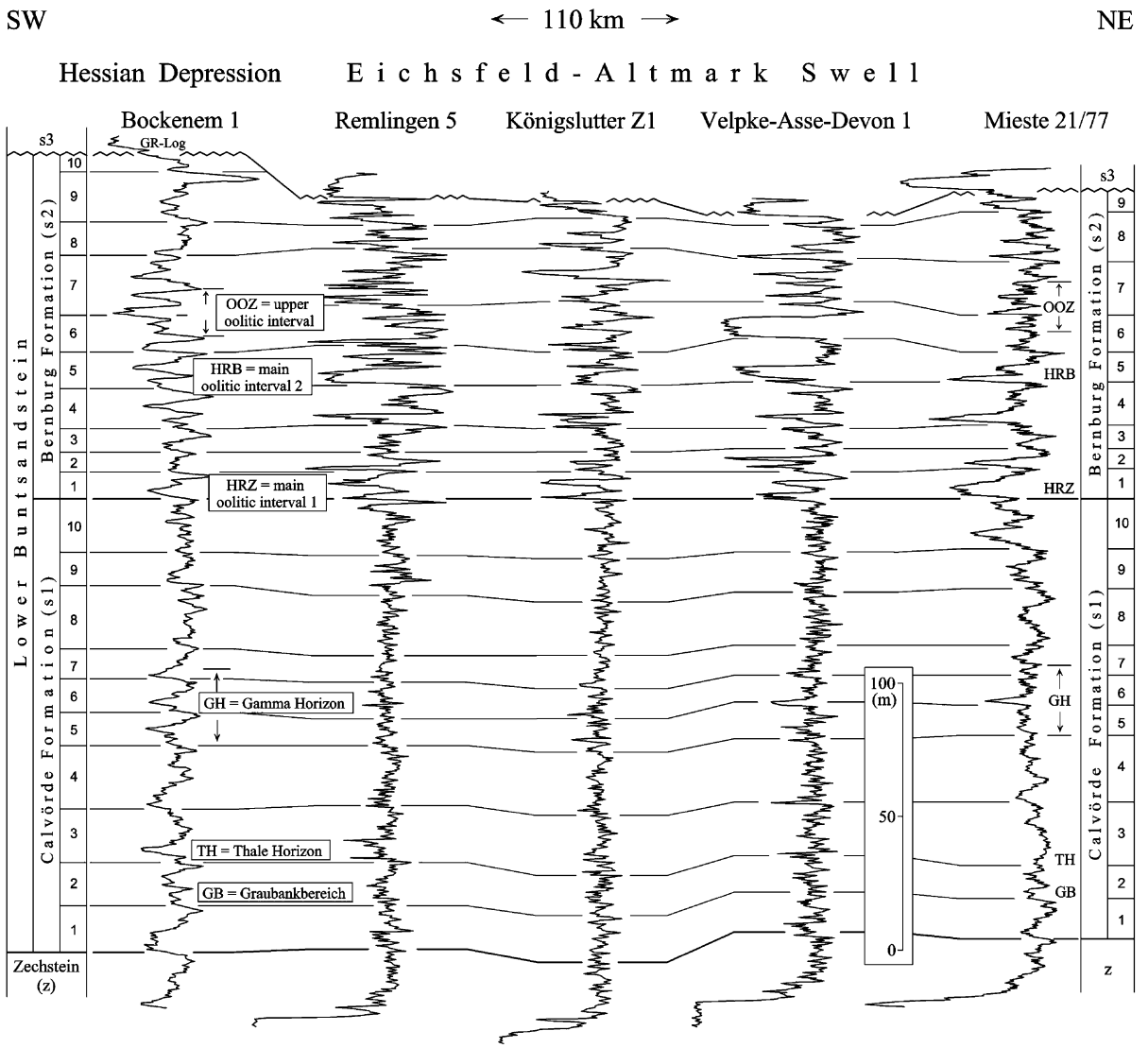


Fig. 4. Well GR log correlation of the Lower Buntsandstein in a SW–NE-trending transect crossing the paleographical elements Hessian Depression and Eichsfeld–Altmark Swell (see also Fig. 1). English terms for displayed marker beds are informal.

tions (IRM) were applied with a 2G Enterprises 660 pulse magnetizer and measured with a Molyneux MiniSpin fluxgate magnetometer (noise level 0.2×10^{-3} A m^{-1}). About 15% of the samples was exposed stepwise, to peak fields of 10–2700 mT along the positive z-axis of the specimen in order to record complete IRM acquisition curves. For a more conclusive interpretation, IRM experiments were combined with subsequent thermal demagnetization of IRM, because different magnetic minerals having similar coercivities generally have different characteristic unblocking temperatures.

4.2. Rock magnetic results

The four diagram pairs in Fig. 5 illustrate the general differences in the magnetic mineral content depending on the lithofacies types. Unweathered limestones contain a low-coercivity mineral,

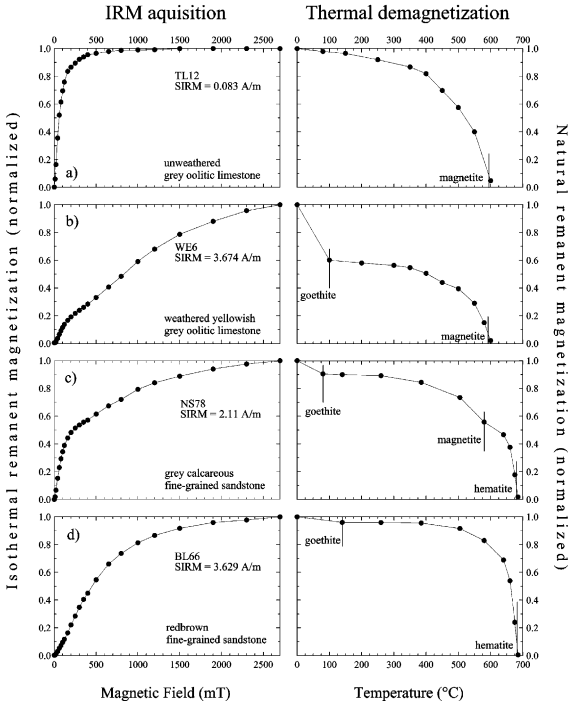


Fig. 5. (a–d) IRM acquisition in fields up to 2700 mT and subsequent stepwise thermal demagnetization of representative lithofacies types from Lower Buntsandstein. Both diagrams show normalized intensities.

acquiring the majority of IRM below 200 mT and unblocked close to 590°C, which can be ascribed to magnetite (Fig. 5a). The weathered limestones show, beside magnetite, a high-coercivity mineral which does not reach saturation at 2700 mT, having an unblocking temperature of ~80°C, identified as probably goethite (Fig. 5b). The gray sandstones in general contain three magnetic minerals (Fig. 5c). Beside goethite and magnetite, there is another mineral with a maximum unblocking temperature close to 680°C, indicating hematite. In the red-brown sandstones, goethite and hematite are recognized from maximum unblocking temperatures of approximately 100°C and 680°C, respectively (Fig. 5d).

4.3. Paleomagnetic results

The NRM intensities range from 0.1 to 21×10^{-3} A m^{-1}, with the red-brown lithologies in general displaying higher intensities than the gray lithologies. The values of the initial susceptibility vary between 10 and 250×10^{-6} (SI). According to results from demagnetization, two types of magnetization were defined.

4.3.1. Type I magnetization

This group is indicated by two superimposed components of magnetization, related to lower and higher unblocking temperature ranges, with significant differences between the red-brown and gray lithologies. Examples of type I magnetizations from red-brown lithologies are shown in Fig. 6a,b. Component A is characterized by northerly declinations and relatively steep positive

Table 1
Mean paleomagnetic directions of the investigated outcrops

Section	N_1, N_2	In situ directions				Tilt-corrected directions				VGP	
		D (°)	I (°)	k	α_{95}	D (°)	I (°)	k	α_{95}	Lat., Long. (°)	dp, dm (°)
Bad Salzdetfurth	115, 096	19.6	73.5	23.6	3.2	11.2	24.2	27.8	3.0	49.6N, 173.0E	1.7, 3.2
Marienburg	151, 108	−15.3	32.3	26.4	3.3	17.9	24.8	30.6	3.1	48.3N, 163.0E	1.8, 3.3
Nelben	046, 042	59.8	62.5	40.0	3.5	17.1	25.6	45.8	3.3	49.4N, 165.7E	1.9, 3.6
Beesenlaublingen	108, 094	21.6	25.0	51.4	2.2	20.6	26.0	57.8	2.1	48.6N, 160.7E	1.2, 2.2
Thale	162, 118	211.7	80.4	24.7	2.7	18.7	28.5	35.6	2.3	50.5N, 161.9E	1.4, 2.5
Westeregeln	046, 033	256.5	59.6	22.4	6.9	12.1	30.7	41.5	5.0	53.3N, 171.7E	3.1, 5.6
Caaschwitz	033, 017	18.5	34.6	80.0	4.0	17.1	32.6	68.2	4.3	54.1N, 163.4E	2.8, 4.9
Lindenschlucht	033, 019	10.5	38.4	21.4	8.5	11.6	24.4	22.9	8.2	50.1N, 173.9E	4.7, 8.7
Overall direction		6.1	62.7	5.3	28.9	16.4	27.5	333.9	3.3	50.6N, 165.6E	2.0, 3.6

N_1, number of specimens demagnetized; N_2, number of specimens used for statistical analysis; D, Declination; I, Inclination; k, precision parameter; α_{95}, radius of cone of 95% confidence about the mean direction; Lat., Long., latitude and longitude of north paleomagnetic pole; dp, dm, semi-axes of the α_{95} confidence oval about the paleomagnetic pole.

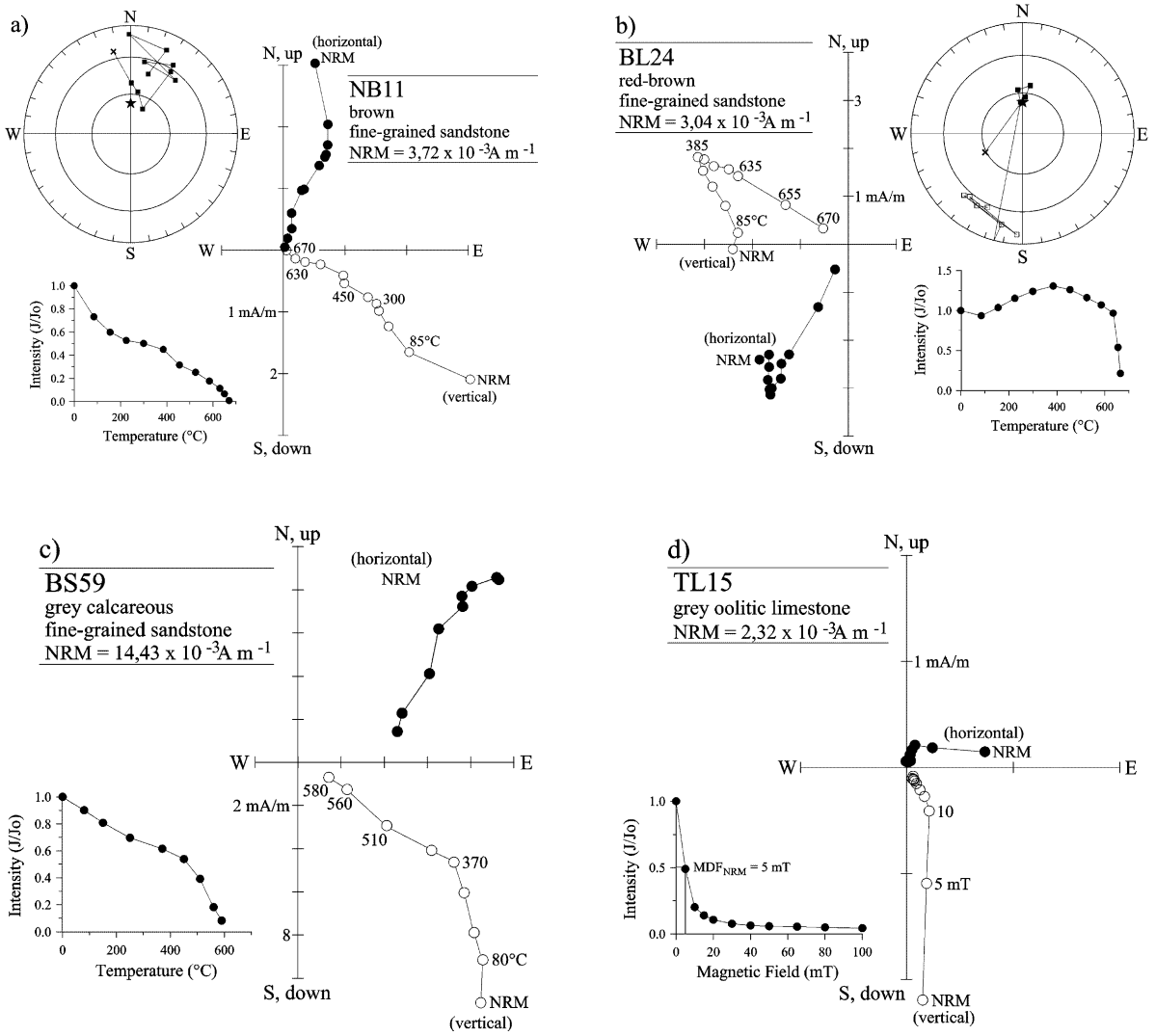


Fig. 6. (a–d) Vector end-point diagrams, stereographic projection of difference vectors and normalized intensity decay curves illustrating thermal and alternating field demagnetization of NRM for four representative samples. Results assigned to type I (panels a–c) and type II (panel d) magnetizations after tectonic correction. Samples are identified with abbreviation of section name and chronological sampling number. The asterisk in the equal-area projection represents the local present-day field direction (PDFD). Demagnetization of type I magnetizations (panels a–c) at first removes a steeply inclined and northerly directed present-day field component A at temperatures below 400°C. Component B, usually converging towards the origin, is isolated over a discrete unblocking temperature range from about 400 to 580°C, indicating magnetite (panel c) and from about 400 to 680°C, indicating hematite (panels a and b), respectively. This direction is regarded as primary magnetization being close to the expected Early Triassic geomagnetic field direction for Central Germany. The results of type II magnetizations (panel d) reveal only a steep present-day field overprint.

inclinations (60°–70°) close to the local present-day geomagnetic field direction ($D = 0°$, $I \approx 66°$). Therefore, this component is ascribed to a recent overprint, which is readily removed below 400°C. The remaining component B, indicated by a maximum unblocking temperature of 680°C, yielded two antipodal groups of directions: (a) northeasterly declinations (Fig. 6a) with shallow positive

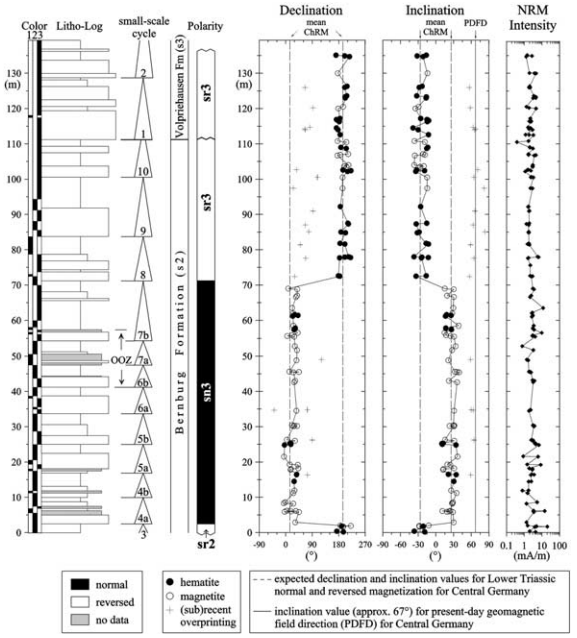


Fig. 7. Magnetostratigraphy of the Bernburg and Volpriehausen formations in the Marienburg section, with inclinations, declinations and NRM intensities versus depth. Type I magnetizations are used to interpret polarity in which ChRM carried by hematite is characterized by solid circles and those carried by magnetite by open circles, respectively. Indicated by crosses are type II magnetizations. See Fig. 3 for key.

inclinations (10°–40°) and (b) southwesterly declinations (Fig. 6b) with shallow negative inclinations (−10° to −40°). In the orthogonal projections this component is revealed as a linear demagnetization trajectory towards the origin. It is regarded as a characteristic magnetization, usually being isolated over an unblocking temperature range from 400°C to 680°C, but most commonly above 600°C, indicating hematite as the carrier of remanence. The mean paleomagnetic direction of component B from outcrop sections ($D = 16.4°$, $I = 27.5°$, $\alpha_{95} = 3.3°$) (Table 1) is close to the expected Early Triassic geomagnetic field direction for Central Germany [42,43]. The characteristic magnetizations pass the reversal test.

The NRM of gray lithologies usually consists of two components (Fig. 6c). First removed with a maximum unblocking temperature below 400°C is a northerly directed relatively steep inclination, which is identified as a recent overprint and referred to as component A. The remaining magnetization shows both polarities: northeasterly declinations with shallow positive inclinations (Fig. 6c) and southwesterly declinations with shallow negative inclinations. This component B, usually converging towards the origin of the orthogonal projections, is isolated over an unblocking temperature range from 400°C to 580°C, but most commonly above 500°C, indicating magnetite. The direction of component B is regarded as a characteristic magnetization being close to the expected Early Triassic geomagnetic field direction for Central Germany. About 79% of all samples belongs to type I magnetizations. They were used to construct a composite magnetic polarity scale.

4.3.2. *Type II magnetization*

About 21% of all samples exhibits, at best, only a single remanence direction, converging towards the origin. This type II magnetization, shown in Fig. 6d, is characterized by steep positive inclination values (50°–70°), which approximate the local present-day field direction. It is readily removed by alternating fields of up to 20 mT and by thermal demagnetization below 400°C. Demagnetization to higher temperatures often yielded scattered directions. Hence, no identifiable characteristic remanence is apparently present in these samples. Thus, type II magnetizations are unsuitable for magnetostratigraphy.

5. Magnetostratigraphy

The stratigraphic plots of the Marienburg and Bad Salzdetfurth sections display the mean declinations, inclinations and NRM intensities of all samples with respect to lithology (Figs. 7 and 8). Both sections cover, together, almost the complete Bernburg Formation. Type I magnetizations indicate one normal (sn3) and two reversed (sr2, sr3) magnetozones. The lowermost part of the Bernburg Formation (cycles 1–3) belongs to sr2. The following magnetozone sn3 encompasses cycles 4–7. In the Bad Salzdetfurth section this interval is dominated by hematite magnetizations, and by magnetite magnetizations in the Marienburg section. The insignificant discrepancy between mag-

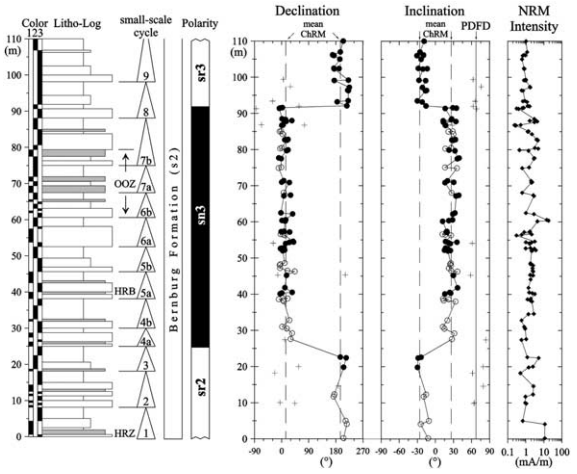


Fig. 8. Magnetostratigraphy of the Bernburg Formation in the Bad Salzdetfurth section. See Figs. 3 and 7 for key.

netostratigraphic and lithostratigraphic data, concerning the sn3–sr3 polarity transition, is due to the fining-upward cycle boundaries, which, in contrast to those of the base-level cycles, are somewhat diachronous. The uppermost Bernburg Formation and the lowermost Volpriehausen Formation belong to magnetozone sr3, comprising the Lower–Middle Buntsandstein boundary. In summary, both outcrops can be correlated in detail using lithostratigraphy, GR logs and magnetostratigraphy (Figs. 7 and 8).

The Schlierbachswald 4 well covers about 400 m of uppermost Zechstein to lowermost Middle Buntsandstein (Fig. 9). Magnetostratigraphic results delineate four normal and five reversed magnetozones. Only the inclinations could be used to determine the polarity, because the core was not horizontally oriented. The uppermost Zechstein comprises one normal (znz) and two reversed (zry, zrz) magnetozones. In the overlying Lower Buntsandstein are delineated three normal (sn1 to sn3) and three reversed (sr1 to sr3) magnetozones. The base of the Buntsandstein is located within the lower third of the thick magnetozone sn1 and the boundary between the Lower and Middle Buntsandstein is in sr3. The results obtained in the Schlierbachswald 4 well (Fig. 9) are consistent with the polarity pattern of the Marienburg and Bad Salzdetfurth outcrop sections (Figs. 7, 8 and 10). The inter-section correlation of all nine investigated profiles allows the establishment of a composite magnetic polarity scale for Central Germany (Fig. 10). The sr2–sn3 polarity transition within the Bernburg Formation is also documented in the Beesenlaublingen and Lindenschlucht sections. The samples from the Thale, Nelben, Caaschwitz and Westeregeln sections display normal polarity from the thick magnetozone sn1.

The preservation of a dual polarity magnetization close to the expected Early Triassic direction, the repetition of a comparable polarity pattern in several sections over a distance of more than 200 km and the lack of polarity changes in the places where magnetite became the magnetic carrier instead of hematite, are the main arguments supporting an early acquisition of the characteristic magnetizations.

6. Global magnetostratigraphic correlation and Permian–Triassic boundary

In order to place the magnetostratigraphy of Central Germany in a broader context, it is compared with Late Permian to Early Triassic polarity records that have been established in Poland [44], the Southern Alps, Italy [7], the Canadian Arctic [4] and Svalbard (Boreal composite) [5] (Fig. 11). Both records of the Germanic Trias are in good agreement. The Lower Buntsandstein and the corresponding Baltic Formation, respectively, consist of three normal and three reversed magnetozones. The base of the Buntsandstein is situated within the lowermost part of a remarkably thick normal magnetozone (sn1, Tbn1) and the Lower–Middle Buntsandstein boundary within a reversed zone (sr3, Tbr3). Magnetostratigraphy indicates the base of the Bernburg Formation to be located within Tbr2 of the Polish Baltic Formation.

Our results are also in good agreement with the polarity pattern from the Southern Alps [7] (Fig. 11). The magnetozones sn1 and sn3 appear to be the equivalent of N2 and N3 of the Tethyan record. The Bellerophon–Werfen Formation boundary approximately correlates with the base of the Upper Fulda Formation. Consequently, the base

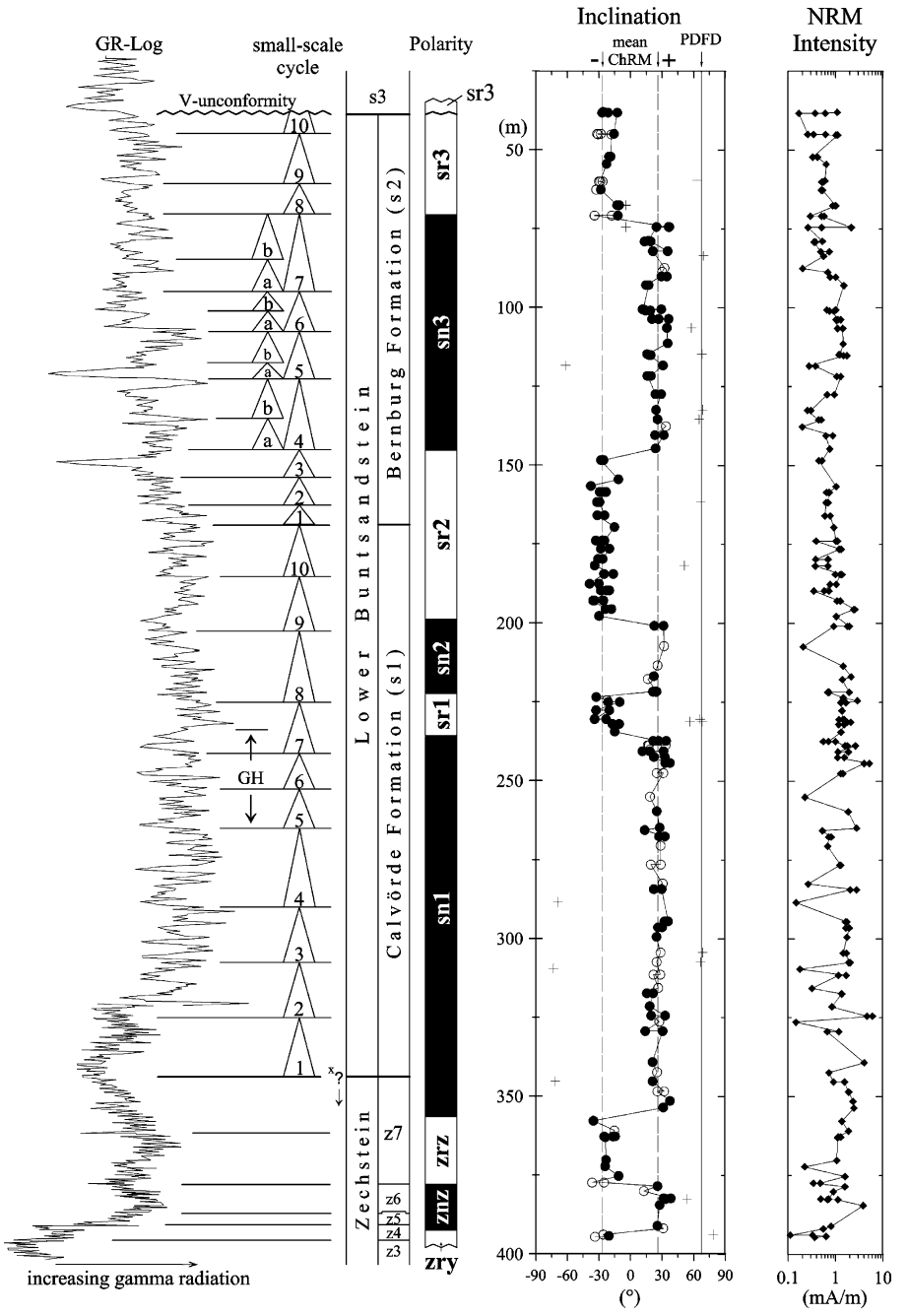


Fig. 9. Magnetostratigraphy of uppermost Zechstein to lowermost Middle Buntsandstein in the Schlierbachswald 4 well (see Fig. 7 for key). As the cores were not oriented with respect to their azimuths, only the inclinations were used to determine the polarity. [x] Zechstein–Buntsandstein boundary according to Käding.

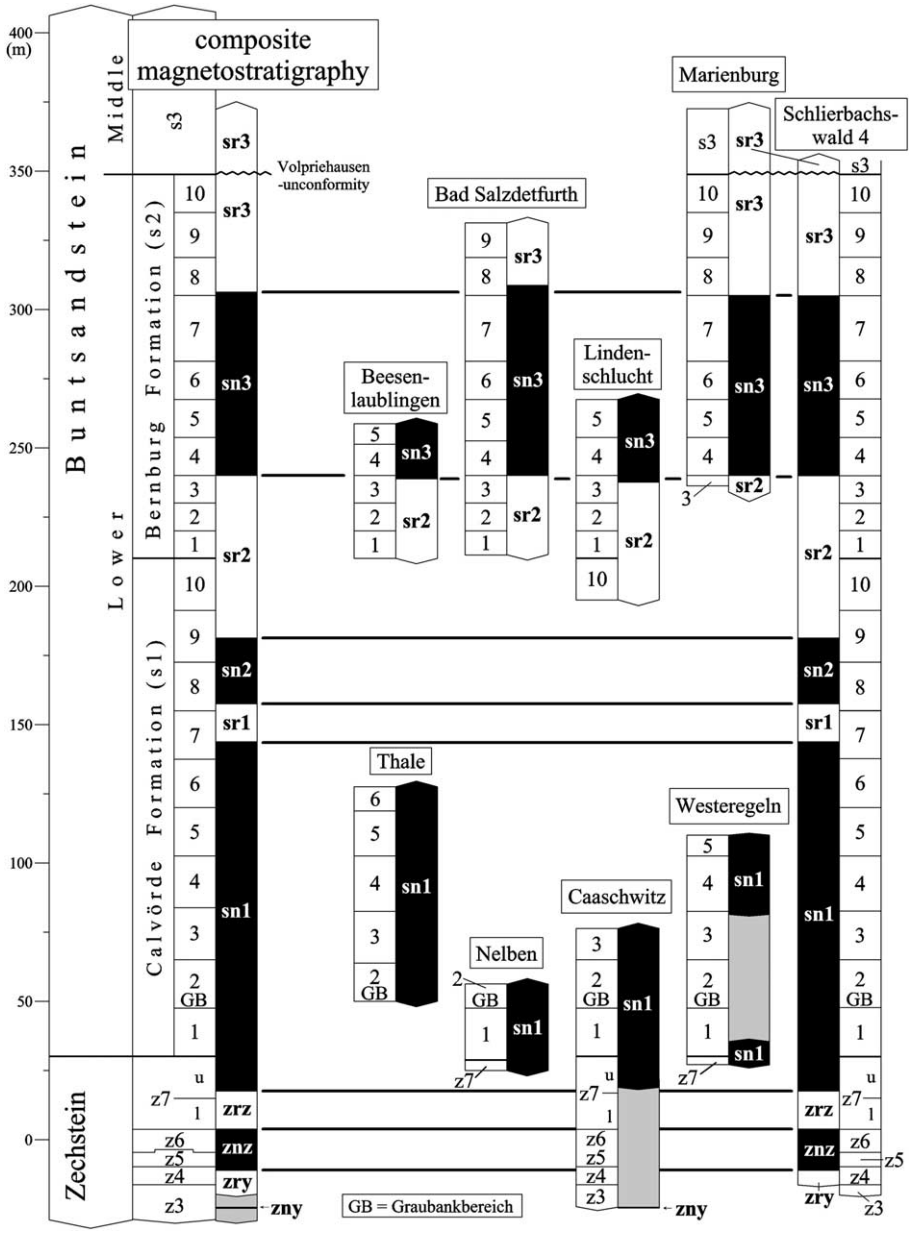


Fig. 10. Composite magnetostratigraphy of uppermost Zechstein and Lower Buntsandstein of Central Germany compiled from nine sections and referred to the high-resolution lithostratigraphy.

of the Bernburg Formation probably lies within the lowermost Seis Member and the base of the Middle Buntsandstein within the uppermost Seis Member. This magnetostratigraphic correlation is in accordance with biostratigraphy [38,39].

By decision of the IUGS the Global Stratotype Section and Point (GSSP) of the PTB is now the Meishan section D (South China), the base of Bed 27c, at the first appearance of the conodont *Hindeodus parvus* [3]. The presence of *H. parvus* in the Bulla section (Southern Alps, Italy) [45] indicates that at least the lower part of the Tesero Oolite

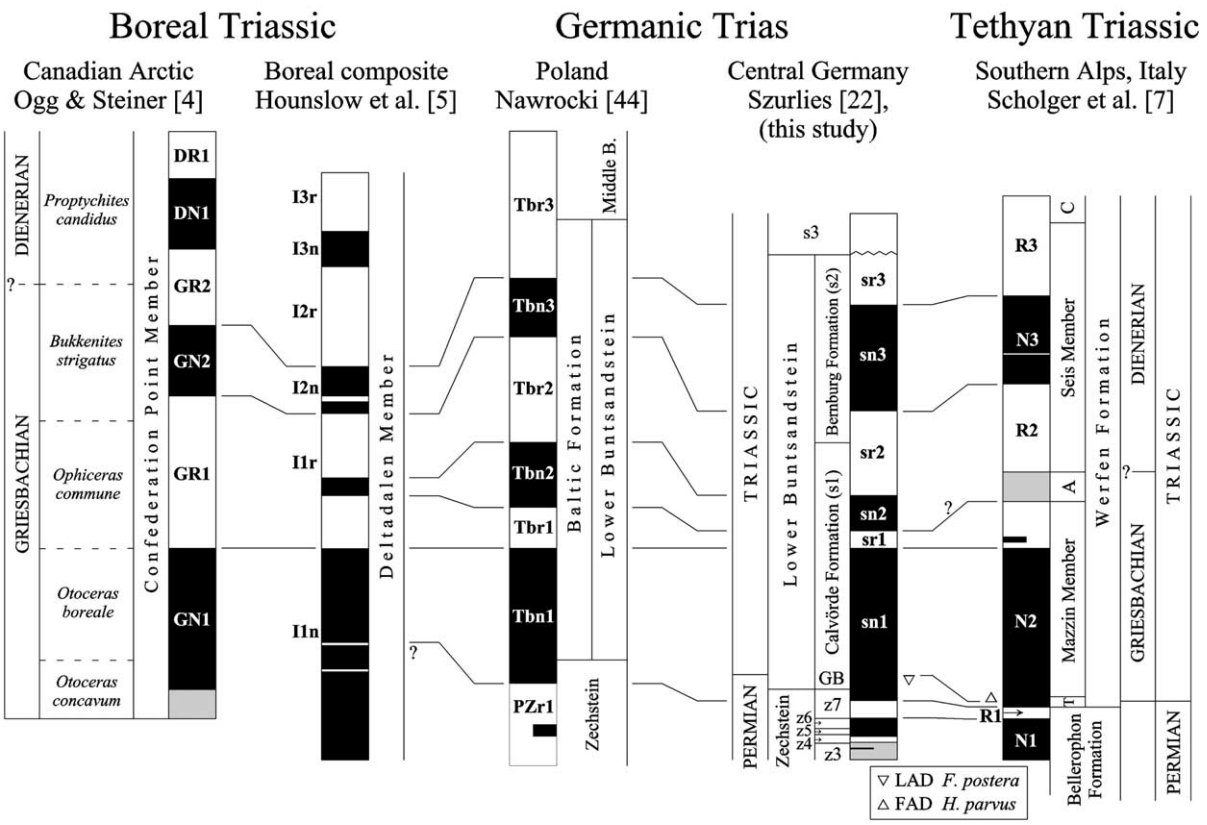


Fig. 11. Magnetostratigraphies of Central Germany [this study] and Poland [44] compared with magnetic records from the Boreal realm of the Canadian Arctic [4] and Svalbard (Boreal composite) [5] as well as from the Tethyan realm of the Southern Alps, Italy [7]. Thickness not to scale. C, Campil Member; A, Andraz Horizon; T, Tesero Oolite Horizon; GB, Graubankbereich.

Horizon is of Late Permian age. Therefore, the PTB in the Southern Alps is situated within the lowermost part of the remarkably thick magnetozone N2, which is correlated with sn1 and Tbn1 of Central Europe (Fig. 11). Hence, it can be concluded that the PTB in Central Germany is situated in the lowermost part of sn1 within the lowermost Buntsandstein. This magnetostratigraphic conclusion is strongly supported by biostratigraphic correlations based on conchostracans and sporomorphs. According to Ecke [35] and Kozur [36,39] the PTB is situated within the 'Graubankbereich', forming the 6–7 m thick basal sandy–oolitic part of cycle 2 of the Calvörde Formation and characterized by gray siltstones and shales. The lowermost part of the Calvörde Formation below the 'Graubankbereich' forms the *Falsisca postera* Zone [39]. It contains the sporomorph association of the *Lundbladispora obsoleta–Lunatisporites noviaulensis* Zone, which can be correlated with the same sporomorph association of the lower Tesero Oolite Horizon of the Southern Alps [38,39], belonging to the *H. praeparvus* Zone. Moreover, the megaspore *Otynisporites eotriassicus*, being common in the *F. postera* Zone of Central Germany and in the lowermost Baltic Formation of Poland, is also present in the Tesero Oolite Horizon [38–39] and in other continental basins [46]. Thus, biostratigraphic and magnetostratigraphic results are consistent, indicating that the PTB in Germany and the Southern Alps is situated in the lowermost part of the thick normal magnetozone sn1 and N2, respectively (Fig. 11). Therefore, in Central Germany the PTB is about 20 m above the Zechstein–Buntsandstein boundary.

Furthermore, this correlation is supported by recent carbon isotope data. The curve of $\delta^{13}C_{org}$ as well as plots of $\delta^{13}C_{carb}$ (Hansen et al., personal communication; Korte, personal communication) show a distinct negative excursion within the 'Graubankbereich', both of which are characteristic for the PTB interval of other marine and continental sections (e.g. [47–49]). Moreover, microspherules (MS) known from several marine sections across the PTB (e.g. [50]) were found in the Nelben section [51] (Fig. 1). The MS maximum frequency interval is below the 'Graubankbereich' within cycle 1 of the Calvörde Formation and the lower *F. postera* Zone, respectively. This interval appears to match closely the maximum frequency interval in the lower *Clarkina meishanensis–H. praeparvus* Zone of the Meishan section [50]. Consequently, carbon isotope data and frequency of MS support the suggestion that the PTB is located within the 'Graubankbereich'.

Also, the available magnetostratigraphies from the Boreal realm [4,5] are in good accordance with the record of Central Germany (Fig. 11). Corresponding to our results, the thickest normal magnetozones are at the base of the Triassic (GN1, I1n). Tozer [52] found *Otoceras concavum* near the base of GN1. He compared the *O. concavum* Zone with the *O. woodwardi* Zone of Perigondwana [53,54]. This opinion was debated among biostratigraphers, with the result that parts of the Boreal Triassic are now considered to be of Permian age. According to Yin et al. [50] and Kozur [38], the PTB is located within the overlying *O. boreale* Zone. Independent of the final placement of the PTB within the Boreal realm, where *H. parvus* is rare, it will certainly be situated within the upper part of the thick magnetozones GN1 and I1n, respectively. Therefore, at least the upper part of GN1 [4] and I1n [5] appears to match closely the magnetozone sn1 of Central Germany. This remarkably thick normal magnetozone is a distinctive feature occurring in virtually all magnetic records across the PTB (Fig. 11; e.g. also [6,8,9,12,13,15]). It forms a good basis for a reliable global magnetostratigraphic comparison. According to our correlation, the magnetozone sn3 is an equivalent of N3 from the Southern Alps and GN2 as well as I2n from the Boreal realm, respectively. This means that the Griesbachian–Dienerian boundary presented by Scholger et al. [7] and Ogg and Steiner [4] is not equivalent. The resulting difference with respect to the position of this boundary is probably the consequence of the poor dating of the stage boundaries of the Dienerian within the Boreal sections studied for magnetostratigraphy [4–6]. According to Ogg and Steiner [4], due to the relatively wide spacing of ammonite-rich horizons and lack of a precise boundary stratotype, especially of the Dienerian stage, the boundary assignments were used only for purposes of magnetozone nomenclature. In the Southern Alps the Griesbachian–Dienerian boundary is considered to be located close to the base of the Seis Member [38]. Independent of this, the magnetic polarity patterns of the Germanic and Boreal realms are in good agreement. The Lower–Middle Buntsandstein boundary is probably located within GR2 and I2r of the Boreal sections.

Unfortunately, it is somewhat problematic to correlate the PTB of the Meishan GSSP by magnetostratigraphic means, because different magnetic records from the Meishan section have been published. According to Li and Wang [12], the PTB is located in the lowermost part of a thick normal magnetozone, which is consistent with results shown in Fig. 11. However, according to more recent paleomagnetic results [17], Bed 27 is characterized by a thin interval of reversed polarity, meaning that the PTB is situated within a reversed interval. This is obviously in contrast to the records of our and other studies. In all, the magnetic polarity scales from China show great inconsistencies. It is notable that in all magnetic records except the recent study of the Meishan section [17] the nowadays *H. parvus*-calibrated PTB is located within the lowermost part of a thick normal magnetozone.

7. Conclusions

The ~300 m thick Lower Buntsandstein of Central Germany can be subdivided into 20 small-scale fining-upward cycles, with 10 each in the Calvörde Formation and the Bernburg For-

mation. These distinctive cycles have been successfully mapped across much of the CEB. The correlation is supported by prominent lithological and gamma markers. Alternatively, these cycles can be interpreted as more or less symmetrical base-level cycles, which are considered to reflect ~100 ka Milankovitch eccentricity cycles. They are believed to provide more or less isochronous units, as suggested by the consistency of magnetozones and lithostratigraphic markers in all intersection comparisons. It can be concluded that the duration of the Lower Buntandstein is at least 2 Ma. According to this assumption, the magnetozones of the lowermost Triassic last ~0.1–0.7 Ma, with an average duration of ~0.3 Ma, which is comparable with the Late Cenozoic magnetozones (e.g. [55]).

The paleomagnetic signal from the Lower Buntsandstein preserves a dual polarity magnetization, which is carried by hematite in the red-brown lithologies and by magnetite in the gray lithologies, respectively. It is allocated to two antipodal groups of directions: northeasterly declinations with shallow positive inclinations (mean values: $D = 16.4°$, $I = 27.5°$) and southwesterly declinations with shallow negative inclinations ($D = 196.4°$, $I = -27.5°$). The characteristic magnetizations pass the reversal test. The appearance of a dual polarity magnetization close to the expected directions for the Early Triassic, the repetition of a comparable polarity pattern in several sections over a distance of more than 200 km, the consistent correlations of lithostratigraphic units and magnetozones, and the lack of polarity changes in the place where magnetite became the magnetic carrier instead of hematite, support an early acquisition of the characteristic magnetizations.

According to our detailed magnetostratigraphic results, the PTB is in the lower part of the remarkably thick magnetozone sn1 of the lowermost Buntsandstein, which is a disinctive feature occurring in virtually all magnetic records across the PTB. By biostratigraphic means the PTB is placed at the base of Oolite Alpha 2 within the 'Graubankbereich' (cycle 2, Calvörde Formation). This correlation is strongly supported by investigations on carbon isotopes and microspherules. Thus, for the first time it can be concluded that the PTB is somewhat more than 0.1 Ma younger than the lithostratigraphic Zechstein–Buntsandstein boundary.

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